



Impact of boundary conditions in a microclimate model on mitigation strategies affecting temperature, relative humidity, and wind speed in a Mediterranean city

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ABSTRACT

This paper investigates several mitigation strategies to detect which has the biggest effect on thermal comfort parameters, namely temperature, relative humidity, and wind speed in a neighborhood in Beirut, the capital city of Lebanon. The strategies include changing the ground surface albedo, replacing old buildings with new ones, and adding vegetation. The simulations, conducted on ENVI-met V4 software, were run on the 20th of January and August representing the two major seasons in the country, winter and summer, respectively. The simulations were carried out on a 200 x 200 x 46-m domain at a resolution of 2 m, which permits the study of small-scale interfaces between buildings, surfaces, and vegetation. The paper then represents various types of lateral boundary conditions (LBC) and illustrates the effects of expanding the area of buildings surrounding the field of interest. The present study will allow urban planners to determine the policies required to ameliorate the urban living environment in cities like Beirut. The simulation exhibits that the greatest impact on thermal comfort parameters is vegetation. It was eminent that the occupancy of vegetation is the most effective mitigation strategy given its capability of enhancing the thermal comfort conditions at the pedestrian's pathway sidewalks. The maximum decrease in air temperature was 2.4 °C at 17:00 on August 20th, while the maximum decrease in the mean radiant temperature was 10.5 °C at 12 at noon on the 20th of August.

1. Introduction

Urban areas accommodate more than half the world's population [1], and account for approximately 80% of the world's energy consumption [2]. Because of rapid urbanization, many cities are becoming crowded and overpopulated, which is contributing to several environmental problems, such as air pollution and decrease in thermal comfort parameters. Additionally, the increasing depletion of green spaces and the high production of anthropogenic heat contributed to the development of the urban heat island effect [3].

In urban centers, meteorological conditions (temperature, wind speed, wind direction, radiation, cloud cover) are affected by several factors. Warm/hot weather and dense urban geometry traps solar radiation and reduces the wind speed causing an increase in air temperature. The rise in the air temperature in such contexts reduces human thermal comfort and consequently drives people to spend more time indoors, which increases the energy consumed for cooling purposes [4]. Another

issue is anthropogenic heat that results from human activities (car emissions, utilization of generators, air conditioning, and industrial facilities) that have a negative impact on the environment [5].

As mentioned, urban geometry modifies solar radiation, wind speed, wind direction and air temperature thus affecting pedestrian thermal comfort. The geometry relates to the height to width ratio (H/W), and the orientation of the street canyons. The materials used for vertical and horizontal surfaces and their albedo properties also affect the thermal comfort parameters. Cities are commonly composed of medium and tall buildings having diverse building materials that significantly affect the urban microclimate. The materials vary from light to dark surfaces having low albedo values [6]. A low albedo, where a small amount of energy is reflected by the surface, causes an increase in the surrounding air temperature due to the heat absorbed and stored in the material. Modifying the performance of building materials allows for better control of the direct and diffused solar radiation emitted into space. As for roads that replace vegetated surfaces; their concrete and asphalt

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materials could also have low albedo properties with high storage capacities, which contribute to the increase of urban temperature.

The absence of trees has a significant effect on the thermal comfort parameters because trees prevent solar energy from reaching particular surfaces, thus maintaining lower temperatures. Since solar radiation is one of the main heat loads in buildings, combining vegetation with the building surfaces' albedo modification could allow controlling the heat gain from incident solar radiation [7]. Moreover, trees' foliage increases the latent heat exchange through evaporative cooling by a process called evapotranspiration, making the air temperature cooler [8]. According to a study, evapotranspiration under optimal conditions can cool the air temperature in the surrounding areas and green spaces by 2–8 °C [9]. Vegetation can also provide other benefits in urban areas such as improving air quality, reducing energy demand for air conditioning, and controlling soil erosion.

A study conducted in Athens shows that the highest temperatures took place in the afternoon hours. It showed that urbanization and anthropogenic heat play a vital role in increasing the urban temperature hours [10]. Another study found that light-colored surfaces are cooler than dark ones when tested on paving materials of different colors, areas, and thermal properties [11]. In another study, 14 types of reflective coatings and finishes of building walls and roofs, sidewalks and pavements were investigated to show that the use of reflective coating with high solar reflectance properties could reduce the surface temperature [12]. Other researchers suggested that urban surfaces should have high albedo properties whenever possible since building materials' specifications affect the adjacent temperature when exposed to solar radiation by transmitting heat to the adjacent environment [13, 14]. The decrease in the ground surface temperature also agrees with other studies whereby increasing the albedo by a factor of 0.25 leads to a 10 °C decrease in the surface temperature [9]. Some studies stated that the increase in the reflected solar radiation exceeded the reduction in longwave radiation, thus negatively affecting the thermal comfort of pedestrians. Cooling roof surfaces will lead to a decrease in their temperature which would result in the reduction of the indoor air temperature and a decrease in cooling energy demand [15]. A study showed that the presence of vegetation lowered both the surface temperature and air temperature near the surface because of evapotranspiration and shading [16]. Increasing the number of trees in urban canyons has a crucial role in enhancing the thermal comfort levels of streets [17]. As per the literature, the proposed measures that affect the temperature, wind speed, and relative humidity include shading of urban areas by trees [18], using cool materials for streets and pavement covers [19], increasing green spaces such as urban gardens or parks [20] and providing water features [21].

To identify the corresponding mitigation strategies, several studies pursued different approaches, based on observational and simulation methods [22]. Observational methods include the field measurement approach (which compares the surface temperature in an urban area with a rural area based on multiple stations), the thermal remote sensing approach (using satellite observations), and the small-scale modeling approach (replacing the urban area with a prototype). The simulation methods, which are based on mathematical models, are continuously improving in terms of accuracy versus cost due to the advancement in computing capacity. Simulation approaches include the energy balance model approach (which provides acceptable accuracy for large-scale energy consumption studies [22]) and computational fluid dynamics (CFD) modeling. With respect to the urban heat island distribution, the latter is more accurate than the energy balance model approach. It is also separated into two different scales: the mesoscale model approach and the microscale models approach, which is the appropriate scale used in this study.

Micro-scale computational modeling has been applied to studies addressing urban heat island, pedestrian thermal comfort, and building energy conservation. It allows investigating the impact of various factors, such as meteorological elements, urban geometry (such as building

orientation and street canyon aspect ratio), and material properties, as well as vegetation.

Microclimate modeling tools such as RayMan and ENVI-met predict climatic features in an urban environment and are crucial for the assessment of the relationship between the climate and the urban scenario alternatives. Moreover, these models can evaluate proposed urban configurations and consequently answer questions relating to design and urban planning.

To make the computational cost of CFD models in unbounded or semi-bounded domains affordable, the domain size is usually restricted [23], which makes the solution highly sensitive to the conditions imposed at the domain boundary. Numerous types of boundary conditions are used in fluid dynamics for various purposes and conditions. In microscale CFD models, the boundary condition resolves the conservation equation inside the surface layer and captures the interaction between the domain and the external environment. This means that the horizontal spatial quantities are simulated with the actual geometry and details with the surface layer interactions. As such, accurate representation of the boundary conditions, especially for the temperature, at the lateral boundaries of the computational domain is one of the main challenges in these models [24]. Numerical implementation of boundary conditions should ideally have the same formal order of accuracy as the field measurements. That is, physically relevant boundary conditions can be optimally synthesized from measurements taken at sites that geographically and temporally represent the domain border appropriately. If the numerical implementation of boundary conditions does not have the same formal order of accuracy as the field measurements, then the computational accuracy will fall in between the simulated values and the field measurement. Thus, the validation of CFD studies for microclimate models is carried out with data from field measurements. A study performed a numerical modeling evaluation for the microclimate of an outdoor urban form in Cairo, Egypt using the regression analysis method and resulted in an R^2 of 0.942 as a coefficient of determination [25]. Another paper discussed the validation process by comparing the measured data with simulated results. The mean absolute error (MAE), root mean square error (RMSE), and the index of agreement (d) were evaluated and resulted in a good agreement for the air temperature (MAE = 1.28 C, RMSE = 1.63 C, and d = 0.99) [26]. The accuracy varies over the field depending on the points near the boundary. This will lead to computational results where points far from the boundary would be near the field accuracy, while the accuracy near the boundary would be close to the numerical accuracy of the boundary conditions [27]. In the absence of accurate knowledge of the spatio-temporal variation of the thermophysical properties at the boundary, specification of the boundary conditions commonly involves assumptions that render them physically weak because they do not account for local variations arising from the heights and geometries of nearby buildings [22]. As an alternative, the multi-scale coupled approach that represents the boundary conditions using measurements that sufficiently resolve the boundary spatially and temporally, and considers the multi-scale nature of the urban atmosphere, has the potential to improve the accuracy of the CFD microscale by providing more accurate boundary conditions [24]. One approach is to employ nested grids, as is commonly done in weather forecasting [28]. In this approach, the boundary conditions at the boundaries of the smaller domain are obtained from the solution in the larger domain. This propagation of information can be one way (from larger domain to smaller domain) or two-way. In addition to being expensive, the option of using nested grids is absent in widely used microclimate modeling tools (e.g., ENVI-met) for simulating mass, momentum, and energy transport in urban environments. To this end, in this paper, we propose to employ a buffer zone around the domain of study. The size of the buffer zone is selected in such a manner that the solution in the domain of study does not change upon a further increase in the buffer size.

Previous versions of ENVI-met allowed the boundary conditions of the simulation to be adjusted by setting some initialization values, such

as air temperature, wind speed and direction, relative humidity, building interior temperature, and the lateral boundary condition that would be used before starting the simulations. This affects the accuracy of the microscale model results because it has a very limited impact on the development of the boundary condition [29]. The new version of ENVI-met V4.4 allows the definition of the diurnal atmospheric boundary conditions that help in achieving a better comparison between a simulated scenario and the measured one [29]. ENVI-met requires meteorological inputs for the initial and boundary conditions that can be obtained from a mesoscale meteorological model or observations at a nearby site [30]. Based on the measured meteorological data, different boundary approaches can be chosen within ENVI-met using either cyclic or full forcing boundary conditions. This allows the definition of the diurnal variations as boundary conditions for the following measured thermal comfort parameters: air temperature, humidity, wind speed, wind direction, and radiation.

In a Mediterranean city like Beirut, the absence of mitigation strategies makes it very important to study, define and enforce climate-sensitive urban planning. Therefore, the objective of this work is to study the impact of implementing mitigation strategies (such as adding vegetation, changing building heights and materials, and modifying the albedo of the ground cover) on the thermal comfort parameters within a neighborhood in Beirut. To the Authors' knowledge, several studies address the effect of boundary conditions and perform the calibration to improve the model's accuracy, as in a study by Detommaso et al. that showed the methods used for calibration and validation of Envi-met models and that many other papers conducted the simulations without validating there models [31]. However, no studies that are concerned with microclimate models in subdomains have investigated the effect of increasing the area of the building on the periphery on the accuracy of the conclusion and results, especially when there is lack of field measurements. Thus, this study aims to identify the appropriate boundary condition input in a simulation that defines strategies that affect the thermal comfort parameters in Beirut.

2. Methodology

An investigation was performed to study the effect of the building's shape and size, vegetation, and materials variables on the meteorological conditions in Beirut. An analysis using a case study design to evaluate which mitigation strategy has the most significant effect on the thermal comfort parameters was conducted.

2.1. Case study area

Lebanon, a country on the eastern coast of the Mediterranean Sea, has an area of 10,452 km² and a total population of 5,988,000 [32]. Recent data shows that approximately 87% of this population lives in urban areas [32]. The Hamra neighborhood in Beirut was chosen for the study. In Beirut (the largest urban agglomeration in the country), vegetated areas cover a small percent of the total urban area while asphalt roads cover about 25% of its urban surface area [33]. Beirut, located at latitude 33.8° N, and longitude 35.5° E, falls within the 'CSA' category of Köppen's climate classification system [34], whereby its climate is characterized by a humid and hot summer and a mild winter. Fig. 1 shows the selected case study area in Hamra. This area was selected because it represents the rest of the city in terms of the design of the streets, the variability in heights of the buildings (mid-rise and low-rise buildings), and the lack of vegetation.

Since this research addresses the effect of the mitigation strategies on particular meteorological conditions in Hamra, the next sections present the control variables that will be implemented, the base case scenario as well as the proposed scenarios. The flowchart in Fig. 2 presents the method and the process of the study. In the beginning, the site and location were selected. The definition of the materials relied on the age of the construction due to the limited data regarding the construction aspects of existing buildings in Beirut. Buildings constructed before 1995 were defined as having single walls and single glazing, while constructions after the year 1995 (referred to as new buildings) would have double walls and double glazing, as compliant with the national building law. As a result, we can single out the components used for construction within each period. Additionally, a survey was conducted in the area to calculate the approximate window to wall ratio (WWR) on each façade of the buildings. This information was important for the definition of building volumes within the simulation model. After completing the base case model, the simulation ran for 24 h on two distinct dates: a summer day (20th August), and a winter day (20th January). These days were selected because the 2018 weather forecasting website indicated that the 20th of January was one of the coldest days of the year and the 20th of August was the hottest day. The base case model was tested several times for calibration and for the selection of the convenient number of nesting cells using the L2 norm method. The L2 norm (Euclidean norm) error is a widely used method for the comparison among the cases when there is a lack of field measurements [36]. The L2 norm method was used to calculate the distance of the vector coordinates from the origin of the vector space; this eventually estimates the errors between any two cases.

Several scenarios were proposed and simulated for Beirut city. The



Fig. 1. The location of the study area within Beirut [35].

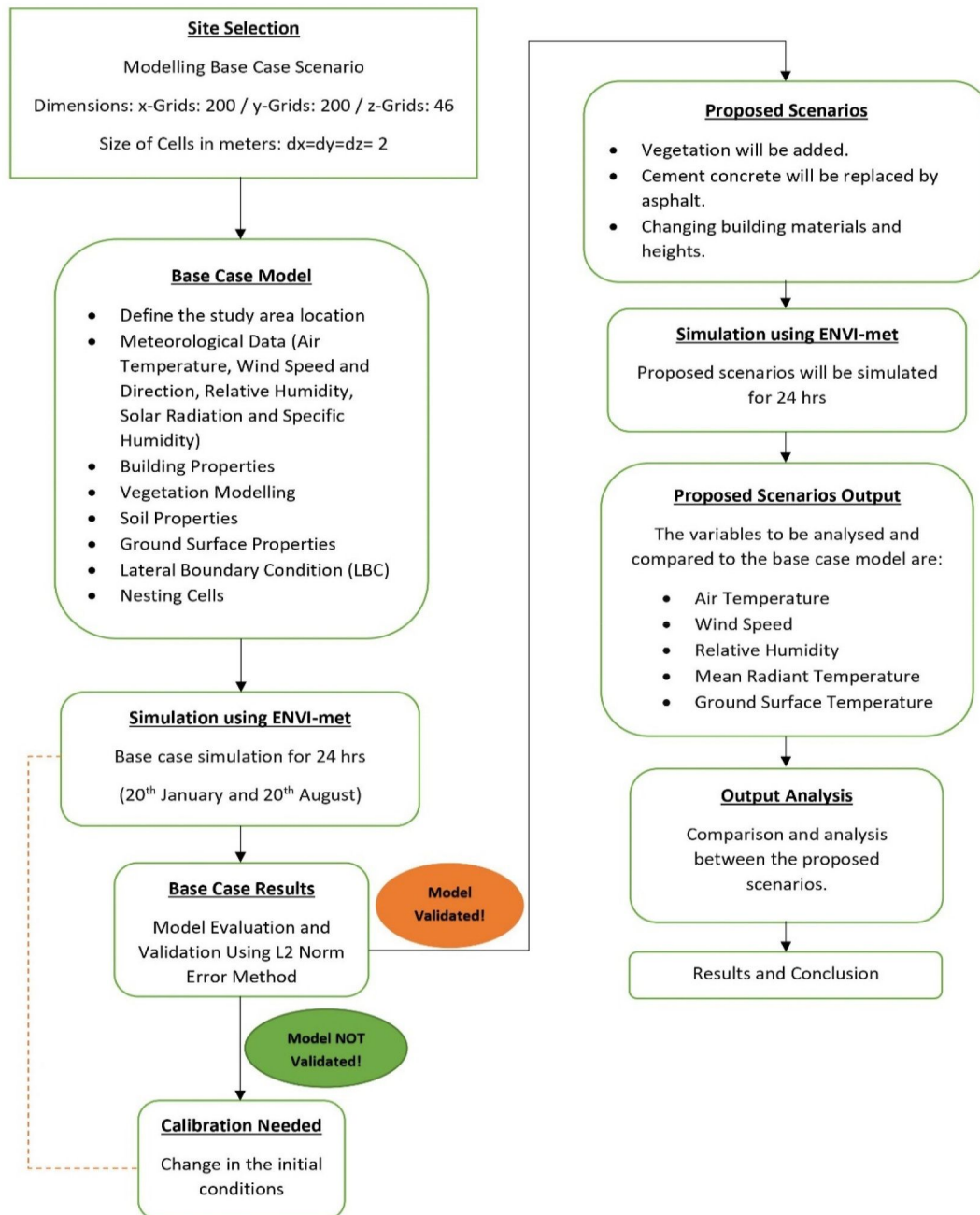


Fig. 2. Flowchart of the study methodology.

selected scenarios were simulated for a 24-h time span, taking into account variables such as air temperature, wind speed, relative humidity, mean radiant temperature, and ground surface temperature. The aim was to compare the base case scenario and the proposed ones. The analysis of the results was based on the location of a pedestrian on a sidewalk near the buildings at an elevation of 1.4 m from ground level, as shown in Fig. 4(b). The height of 1.4 m, representing the center of gravity of a standing person, was chosen because it is the nearest value to 1.2 m on ENVI-met [37]. This allows a better understanding of the effect of implementing mitigation strategies on these variables. The results compared the average hourly data of the pedestrian between the base case and the proposed scenarios; they also compared their absolute difference at noon.

2.2. Control variables

Table 1 presents the three main control variables selected for this study. Variable 1, the ground cover, has been selected to study the various kinds of covers that result in improving the thermal characteristics and reducing the temperature, due to the increase in the albedo. The second variable, vegetation, was chosen due to its cooling effect and contribution to enhancing the thermal conditions in urban spaces. The evaporative cooling process that occurs as a result of converting sensible heat to latent heat facilitates the dissipation of the trapped heat in any urban geometry [38]. The third variable involves replacing old buildings with new ones. The survey that was conducted in the studied area allowed the definition of old constructions (pre-1995) and new buildings that were constructed using a double wall of two layers of masonry concrete units separated by a 5 mm air gap and having double-glazed

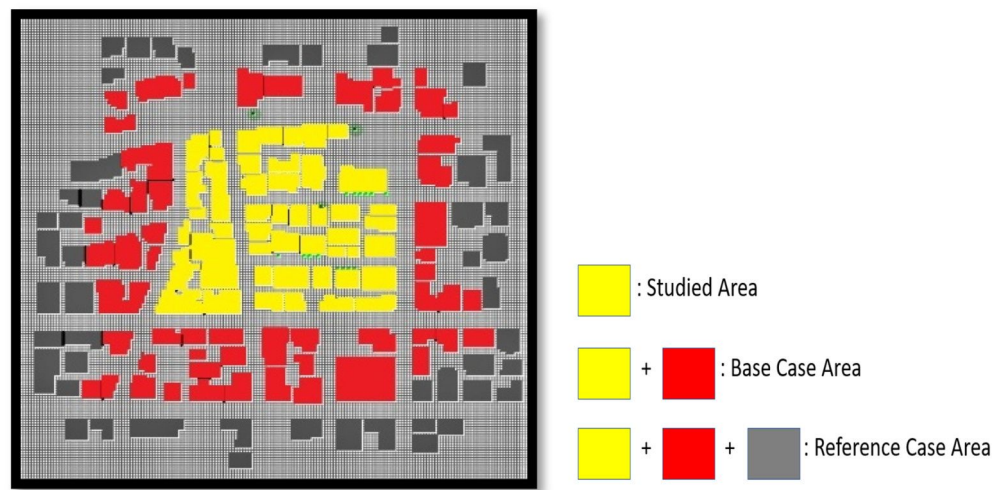


Fig. 3. Input file drawing for ENVI-met simulation.

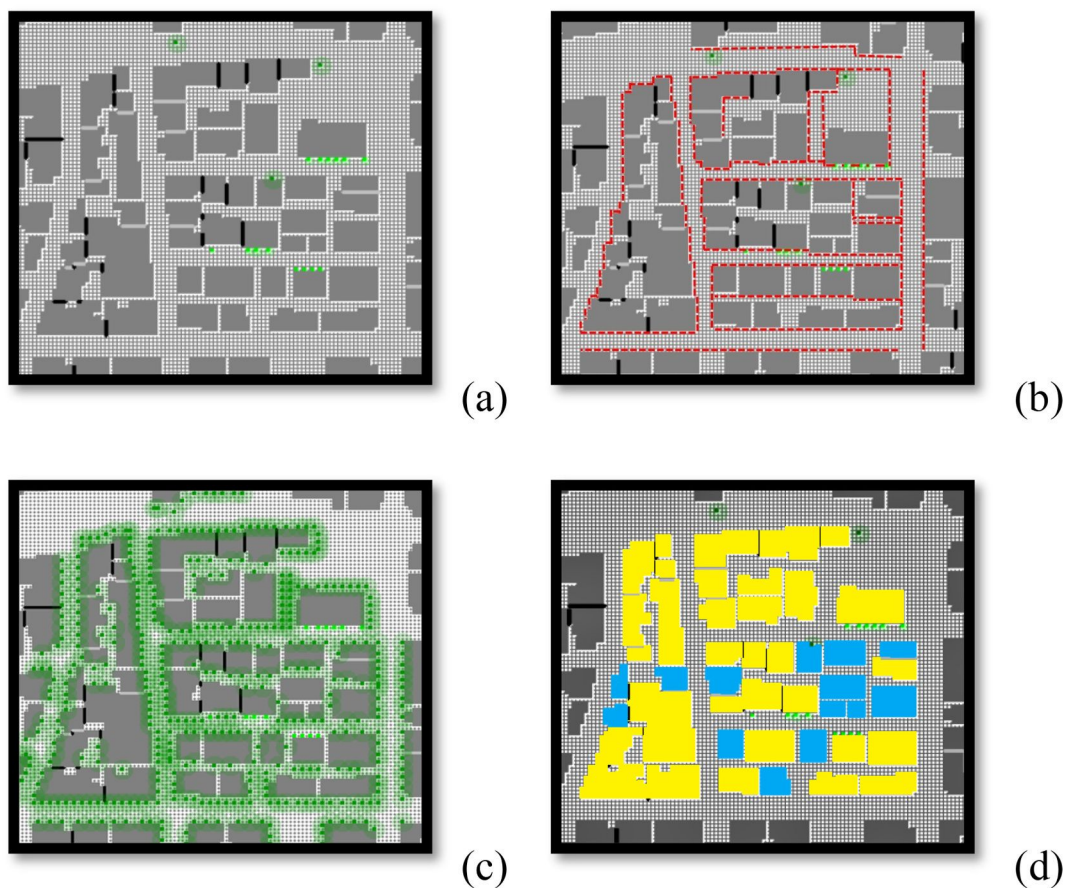


Fig. 4. (a)Base case scenario (BCS), (b)Pedestrian sidewalk, (c)Vegetation Scenario (VS), (d)New Building Scenario (NBS), which are in blue color. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

windows. The number of Xs was added in the table to show the relationship between the variables and the assessment criteria chosen. A single X means that there is a weak relationship between the variable and the assessment criteria, while XX means that there is a strong relationship between them.

2.3. Base case scenario

The base case scenario was simulated using ENVI-met V4 because of its ability to reproduce the atmospheric processes that physically affect the microclimate. This holistic microclimate model couple's thermodynamics and fluid dynamics in complex urban geometry and simulates the surface-plant-air interactions in an urban environment with a typical resolution down to 0.5 m in space and 1–5 s in time [39]. ENVI-met is

Table 1
Control variables with their assessment criteria.

Assessment criteria	Variables		
	Ground Cover	Vegetation	Replacing Old Buildings with New Ones.
Air Temperature	X X	X X	X
Wind Speed	X	X X	X X
Relative Humidity	X	X X	X
Mean Radiant Temperature	X X	X X	X X
Ground Surface Temperature	X X	X X	X

widely considered to be one of the most accurate models that can support understanding the impact of the environment on the microclimate [40]. For example, a study compared three areas in Phoenix and found that the difference between the measured and modeled temperatures was minimal ($\Delta = 0.4^\circ\text{C}$) [41]. The user starts by modeling the scenario, via site selection and defining the number of grids in x, y, and z directions and the resolution which is the size of the cell as an (.INX) file. Then, model configuration and initialization parameters should be input by the user as a simulation file (.SIMX). These include the starting time of the simulation and for how many hours it will last, the air temperature, wind speed measured at 10 m height and direction, relative humidity at 2 m height, building interior temperature, the albedo of walls/roofs, and the heat transmission of walls/roofs. Choosing the appropriate lateral boundary condition and nesting cells is important and depends on the studied area as discussed in section 3. ALBERO is a toolbox where trees and grass databases are built inside the software to check the plant geometry, basic and advanced properties, and root settings. In the database manager, the user can choose and adjust the types of walls and material properties, soils, and ground material. Leonardo is where the software analyzes the model results and creates two and three-dimensional representations and animations. The calculations of the microclimate model include short and long-wave radiation fluxes, evapotranspiration and sensible heat fluxes, dynamic calculation of the surfaces and wall temperatures, the dispersion of gases and particles, and biometeorological metrics such as mean radiant temperature and physiological equivalent temperature.

For this study, the layout of the chosen neighborhood was developed based on a satellite image obtained from Google Earth Pro software and field measurements. The completed area input file was 200 grids in the x and y directions, while 46 grids were considered for the z-direction. Each grid cell in the area input file was set to be 2 m by 2 m in the x-y plane and 2 m in the z-direction. In addition, a border of 15 nesting grids surrounded the simulation area for numerical stability. Regarding the tree species, since the trees in the studied area were not found in the database of the ALBERO toolbox in ENVI-met, a study about the local trees was conducted to build them in the software and use them in the simulations. The study was based on site visits and empirical observations regarding the characteristics of the available trees in the case study area. The study included the type of trees, height, geometrical and foliage characteristics. To create a tree in ALBERO, the height and the width of the plant, the leaf density area (LAD), the maximum depth of the roots as well as the root area density (RAD) were required.

Tables 2 and 3 present the input parameters required for the simulation and the road material characteristics. The surface finishes were set to be the default soil type, while the roads were defined as having asphalt as a material. The 24-h simulation on ENVI-met was initiated at 7:00 a.m. simulating and providing hourly output for the air temperature, wind speed, relative humidity, mean radiant temperature, and ground surface temperature.

2.3.1. Input parameters

Table 2
Model configuration and initialization parameters for the simulations.

	January 20, 2018	August 20, 2018
Starting time	7:00 a.m. (24 h)	7:00 a.m. (24 h)
Input Area	200 x 200 x 46	200 x 200 x 46
Resolution (meters)	2 x 2 x 2	2 x 2 x 2
Air Temperature	12 °C	32 °C
Wind Speed	2 m/s	1.3 m/s
Wind Direction	170 (South-East)	220 (South-West)
Relative Humidity	60%	65%
Nesting cells	15 Grids	15 Grids
Lateral Boundary Condition	Cyclic	Cyclic
Building Interior Temperature	23 °C	21 °C

Table 3
Road material characteristics for the simulations.

	Surface Albedo	Thickness (m)	Volumetric Heat Capacity (J/m ³ . K)
Asphalt Road	0.2	0.3	2.25
Cool Pavement	0.7	0.2	2.08

3. Lateral boundary conditions (LBCs) and nesting cells

This section presents the three possible LBCs proposed in the ENVI-met software and describes the LBC chosen for the study. In addition, nesting cells and the benefit of including them in the study are also discussed.

3.1. Lateral boundary condition

To configure turbulence at the boundaries of the model, ENVI-met provides three types of LBCs, namely “Closed (or Forced) Boundary Conditions”, “Open Boundary Conditions” and “Cyclic Boundary Conditions”. [39].

In the ‘Closed (or Forced) Boundary Conditions’, the values of the one-dimensional model are copied to the border. This approach allows forcing the air temperature and the relative humidity through the diurnal cycle along with the simulation. Moreover, Forced LBC is the most stable condition as it uses the 1-D model to obtain the boundary values, which stabilizes the 3-D model.

In the “Open Boundary Conditions”, the values of the inner points are copied back to the lateral inflow boundary for each time step. This approach is such that the values are copied from the inner parts of the model to the boundaries with the minimum effect of the model boundary onto the inner parts of the model.

As for the “Cyclic Boundary Conditions”, the values of the downstream model border are copied to the upstream model border. This approach assumes the condition that the average conditions of the upstream conditions are comparable to the model area [39,42].

The Open and Cyclic lateral boundary conditions differ from the Forced boundary condition by allowing the simulation to start with few initial parameters. Choosing the appropriate lateral boundary condition depends on the model and the defined methodology.

Fig. 3 represents the input file drawing for the ENVI-met simulation. The yellow-colored buildings display the studied area which is essentially the area of interest with an area dimension of 200 × 200 m. The red-colored buildings were added at the periphery of the studied area for numerical stability; they illustrate the base case area. Since no meteorological data was measured on-site due to the inexistence of weather stations in the studied area, an outer ring of buildings, represented by the grey color, was defined to create the reference case that will make the studied area stable. A study showed that it is crucial to include all the geometric details for a higher resolution, and it is also necessary to add parts of the surroundings in order to obtain adequate results for the area

of interest [43]. Consequently, the reference case area and the studied area are compared to each other using the L2 norm method.

Simulations having both Forced and Cyclic LBCs were conducted to assess which LBC is most suitable for the studied area only. For the Forced LBC and, as mentioned, due to the lack of weather stations in the studied area, the meteorological data was added on an hourly basis from the weather underground website. As for the Cyclic LBC, only initial input was needed at the beginning of the simulation.

3.2. Nesting cells

Nesting cells are grids that surround the simulated area to maintain the numerical stability of the chosen boundary. In the computational domain, nesting grids were added to create a zone in both x and y directions, thus reducing the numerical instability by moving the model boundary away from the studied area.

To investigate the combined effect of adding a building layer at the periphery of the studied area and that of nesting cells, several case studies were tested. Table 4 shows the two reference cases A and B that were selected, consisting of the same base case with an extra periphery layer of buildings but with a different number of nesting cells. Cases 1 and 2 were the studied area without including any periphery buildings with nesting cells equal to 15 and 5 respectively. Cases 3, 4, and 5 included the addition of the periphery buildings onto the base case. Cases 1 to 5 were compared for both reference cases, and the assessment parameters were based on air temperature and wind speed. Both air temperature and wind speed were chosen because they can be represented as a vector form serving to validate the results.

Comparing the results showed that although the same number of nesting cells were used, there was a difference between the two LBCs (forcing or cyclic). The periphery area of buildings was added to investigate its effect on the results of the studied area. The results revealed that both LBCs led to a minimal difference in the air temperature and no difference in the wind speed results.

These simulations illustrated the importance of adding the peripheral buildings on the outer border of the studied area, which lead to numerical stability as compared to the simulation of the studied area alone. Therefore, the Cyclic lateral boundary condition was adopted since it represents a model where the neighborhood has, on average, the same structure as the model area [44,45].

4. Proposed scenarios and assessment parameters

4.1. Proposed scenarios

The base case and the simulated scenarios are listed in Table 5. The first scenario proposed replacing the asphalt cover with a cement concrete cover that has a higher surface albedo. The albedo of the cool pavement is 0.7 (Table 3), whereas that of the asphalt road is 0.2. The second scenario (Fig. 4(c)) increased the cover of vegetation (type: Ficus Elastica) in the total area by 25%. Ficus Elastica is a broadleaf evergreen tree, with a height of 18 m, foliage albedo 0.31, and LAD value 2, and is

Table 4
Different model areas with different numbers of nesting cells.

Cases	Model Area Type	Nesting Cells
Reference Case A	Base Case with an extra periphery layer of buildings	15
Reference Case B	Base Case with an extra periphery layer of buildings	5
Case 1	No Periphery Buildings were added	15
Case 2	No Periphery Buildings were added	5
Case 3	Base Case with Periphery Buildings	15
Case 4	Base Case with Periphery Buildings	5
Case 5	Base Case with Periphery Buildings	1

Table 5

Proposed scenarios with their control variables.

	Ground Cover	Vegetation Cover of the Total Area (%)	Building Heights and Materials
Base Case Scenario (BCS)	Asphalt Cover	1%	• Single & Double wall buildings • Single and Double glass windows
Cool Pavement Scenario (CPS)	Cement Concrete Cover	1%	• Single & Double wall buildings • Single and Double glass windows
Vegetation Scenario (VS)	Asphalt Cover	26%	• Single & Double wall buildings • Single and Double glass windows
New Building Scenario (NBS)	Asphalt Cover	1%	• Double-wall buildings • Double glass windows
Combine Case Scenario (CCS)	Cement Concrete Cover	26%	• Double-wall buildings • Double glass windows

planted on the sidewalks to provide shade and high albedo from the foliage. For the third scenario, old buildings, based on their construction method and height, were replaced with new mid to high-rise developments as allowed by the local building law (Fig. 4(d)). In the fourth scenario, all the scenarios will be simulated simultaneously in order to investigate the combined effect of all the parameters.

4.2. Assessment parameters

As mentioned in Table 1, air temperature, wind speed, relative humidity, and mean radiant temperature were addressed in this study.

Air temperature (T_a) was used to test the temperature variation and distribution due to building arrangements, the building enclosure material properties, and the ground surface materials. Canyons, where the street is surrounded by buildings on both sides, have a different orientation and different exposure to incident solar radiation depending on the varying solar azimuth and altitude. An experimental study established a correlation between the air temperatures compared to solar radiation and noticed that a change in 59 W/m^2 in solar radiation will lead to a change of 1°C in air temperature [46].

Wind velocity was used to evaluate the air movement distribution in the space between buildings. Wind speed affects convection and evaporation and its distribution speed is highly influenced by the orientation of the canyons. Also, during winter, the outdoor wind velocity may create discomfort, which makes it unpleasant for the pedestrians [47].

Relative humidity (RH) was used to assess the effect of the vegetation. The value of the relative humidity has a direct influence on the latent load, consequently on human sweating. Relative humidity, which is higher than 65% during the warm/hot and humid summers results in thermal discomfort for humans [47].

Finally, the mean radiant temperature (T_{mrt}) is defined by ASHRAE as the 'uniform temperature of an imaginary enclosure in which the radiant heat transfer from the human body equals the radiant heat transfer in the actual non-uniform enclosure.' Mean radiant temperature is an important variable that affects outdoor human thermal comfort [48]. In public outdoor spaces, the exposure of the built surfaces (building materials, street pavements, etc.) to direct solar radiation increases their temperature leading to the increase in the mean radiant temperature. A study revealed that adding trees in a park will result in a cooling effect for the pedestrians by decreasing the air temperature and the mean radiant temperature [49].

5. Results

This section displays the results as they relate to the LBCs and the nesting cells for the base case scenario (BCS) simulation, and the other scenarios that were conducted: the cool pavement scenario (CPS), the new buildings scenario (NBS), the vegetation scenario (VS) and the combined case scenario (CCS).

5.1. Parametric analysis using L2 norm method

Using the L2 norm error, the five cases were assessed and compared with reference cases A and B. Fig. 5(a) and (b), for both air temperature and wind speed, show the comparison between the reference case and other cases using L2 norm error. The compared T_{air} and WS values refer to an average air temperature and average wind speed among the pedestrian and sidewalk areas. Cases 1 and 2 showed the maximum error percentage among the five cases. Cases 3, 4, and 5 showed the least percentage error compared to cases 1 and 2. Case 3, which consisted of the periphery buildings outside the studied area, showed minimal error compared to the reference case B which has 0.26% error for the air temperature and 0.75% error for the wind speed.

Also, when comparing case 1 and case 3 that have the same number of nesting cells, the errors for the air temperature and wind speed relative to the reference case A decreased from 7.48% to 0.69% and 41.58%–1.06%, respectively after adding the periphery building layer in case 3. Comparing cases 3, 4 and 5 that have the same model area type but a different number of nesting cells, resulted in showing that case 3 (15 nesting cells) has the least error percentage with respect to the reference cases.

This comparison illustrates the importance of expanding the study area by including buildings at the periphery of this area in order to reach numerical stability (as compared to simulating the base case only, without adding the layer of buildings). Also, the comparison demonstrates that the nesting cells maintain numerical stability, having less effect than that of adding the peripheral layer of buildings.

The following results present the comparison between the BCS and other mitigation scenarios at 1.4 m height for air temperature (T_a), wind speed, relative humidity, mean radiant temperature (T_{mrt}), and the ground surface temperature (T_g) on summer (20th of August) and a winter day (20th of January).

5.2. Air temperature

As per Fig. 6 (a) all the cases have lower T_a with no difference between the BCS and the NBS. The maximum decrease of 2.6 °C in the T_a was at 17:00 in the CCS as compared to the BCS.

Fig. 6 (b) which refers to a winter day, shows that there is no difference in T_a between the BCS and the CPS, while the VS and the CCS show a difference of 0.9 °C compared to the BCS during the daytime and a decrease to 0.5 °C during the nighttime.

Fig. 7 shows the absolute difference in the air temperature between the BCS and various mitigation strategies at noon in addition to the solar orientation and wind direction. There is a slight difference in Fig. 7 (a) compared to the BCS, while the maximum difference is seen in the VS and the CCS Fig. 7 (c, d). These results regarding the decrease in air temperature for both the cool pavement and vegetation scenarios are aligned with results as aforementioned in the literature review [9].

5.3. Wind speed

Fig. 8 shows the results of the ENVI-met simulation for the average wind speed difference among the cases with respect to the BCS on the 20th of August and the 20th of January. In Fig. 8 (a), the mean wind speed at the pedestrian sidewalk for all cases is higher except for the NBS. Moreover, there is a very slight difference between the CPS and the BCS.

Fig. 8 (b) refers to the winter day and shows no difference between the BCS, CPS, and NBS, whereas there is a 0.15 m/s difference between the BCS and the VS and CCS.

Fig. 9 shows the absolute difference in wind speed between the proposed scenarios and the BCS at noon during the summer day. Fig. 9 (c,d) shows a difference of 0.5 m/s in particular areas when comparing the scenarios with the BCS, while Fig. 9 (a) shows a smaller difference (0.1 m/s to 0.2 m/s) between the NBS and the BCS. As for the CPS in Fig. 9 (b), there is no difference compared to the BCS.

5.4. Relative humidity

Fig. 10 (a) refers to relative humidity at 1.4 m height for the 20th of August and shows that the CPS and the VS have higher relative humidity compared to the BCS, while there is no difference between the NBS and the BCS. The largest difference is between the CCS and the BCS reaching 18% during the daytime and decreasing to 11% during the nighttime hours.

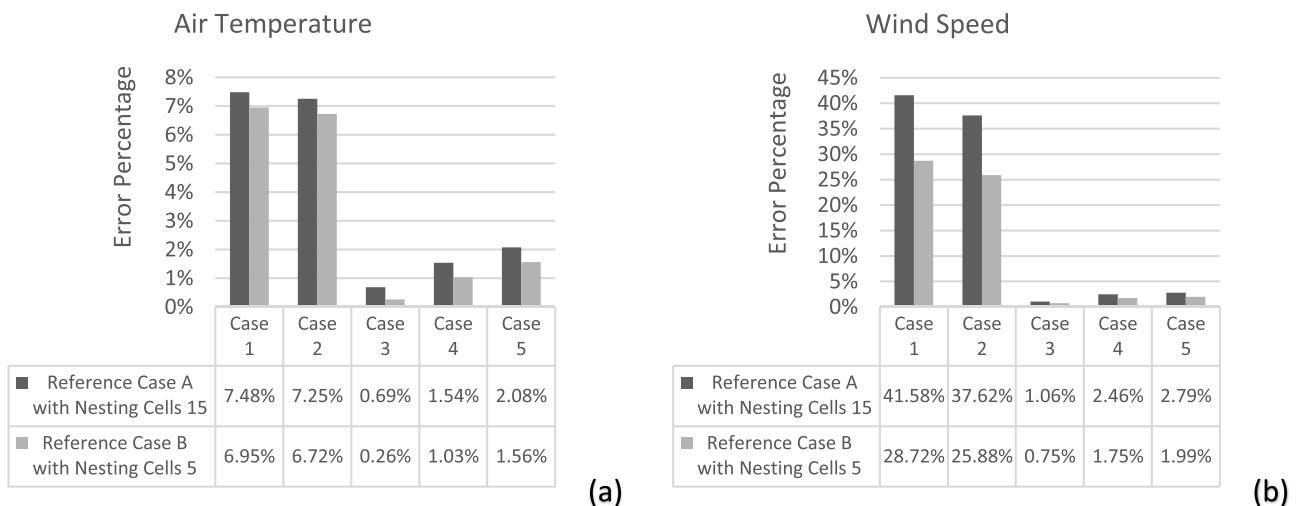


Fig. 5. Comparison between Reference and other cases using L2 norm error.

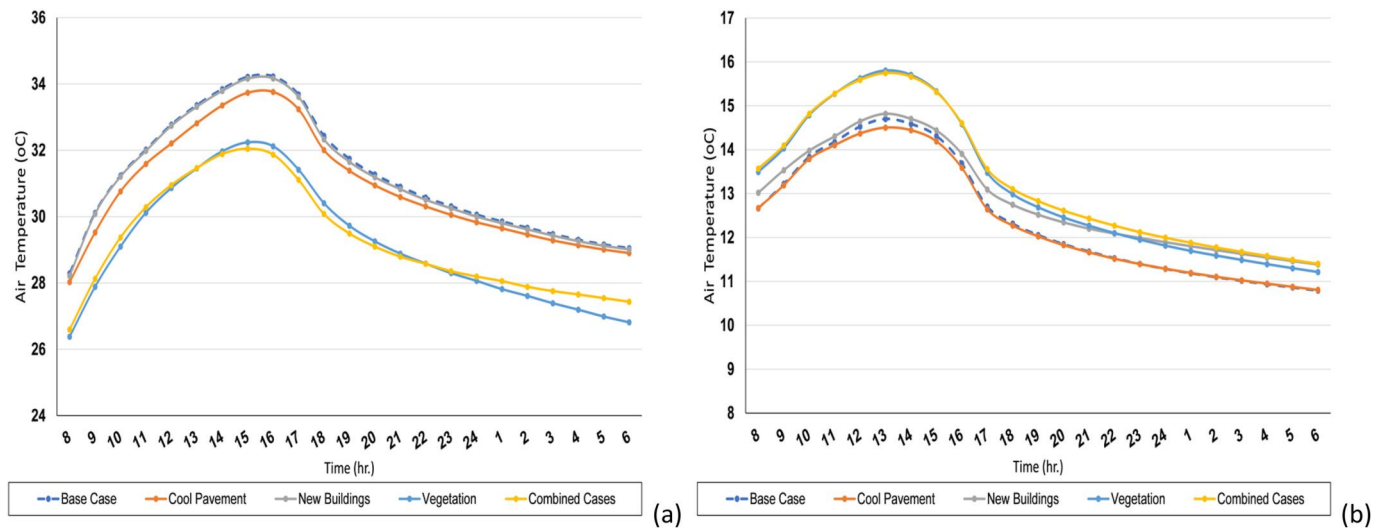


Fig. 6. Comparison of the results for average air temperature between the base case scenario and other mitigation scenarios during (a) 20th August and (b) January 20, 2018 for 24 h.

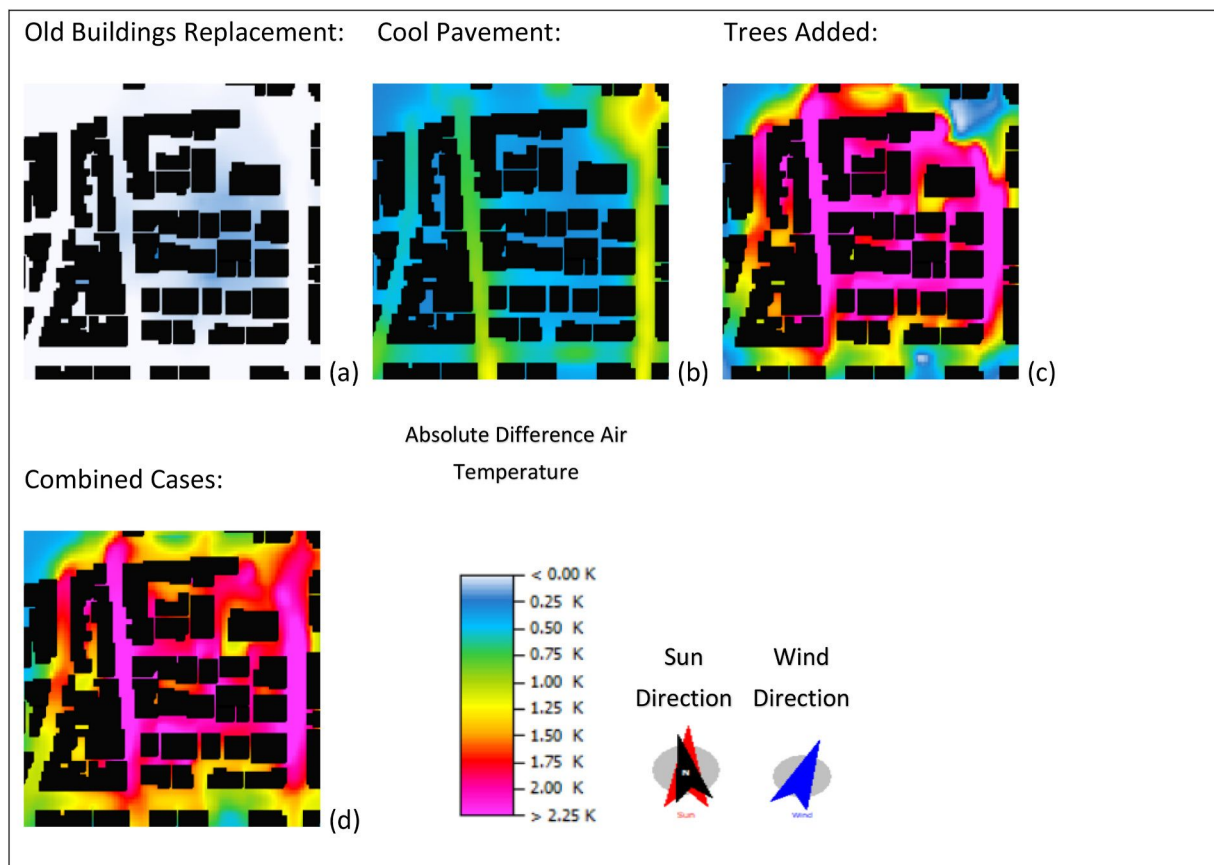


Fig. 7. Absolute difference in air temperature between various heat mitigation strategies compared to the base case scenario at 12:00 at 1.4 m elevation from ground level.

As for Fig. 10 (b), presenting the results for the 20th of January, relative humidity is increasing during the night hours. Throughout the day, the relative humidity in CCS increased by 4.7% with respect to the BCS while it decreased by 3% during the night hours.

Fig. 11 shows the absolute difference in the relative humidity between the BCS and other mitigation strategies at noon. Fig. 11 (a) suggests that the NBS has no impact on the relative humidity. Adding cool

pavement (CPS) Fig. 11 (b) shows a slight increase in the relative humidity compared to the BCS. The simulation results show that the vegetation scenario has the highest impact on relative humidity due to the increase in the rate of evapotranspiration after sunrise, which totally agrees with various studies [50].

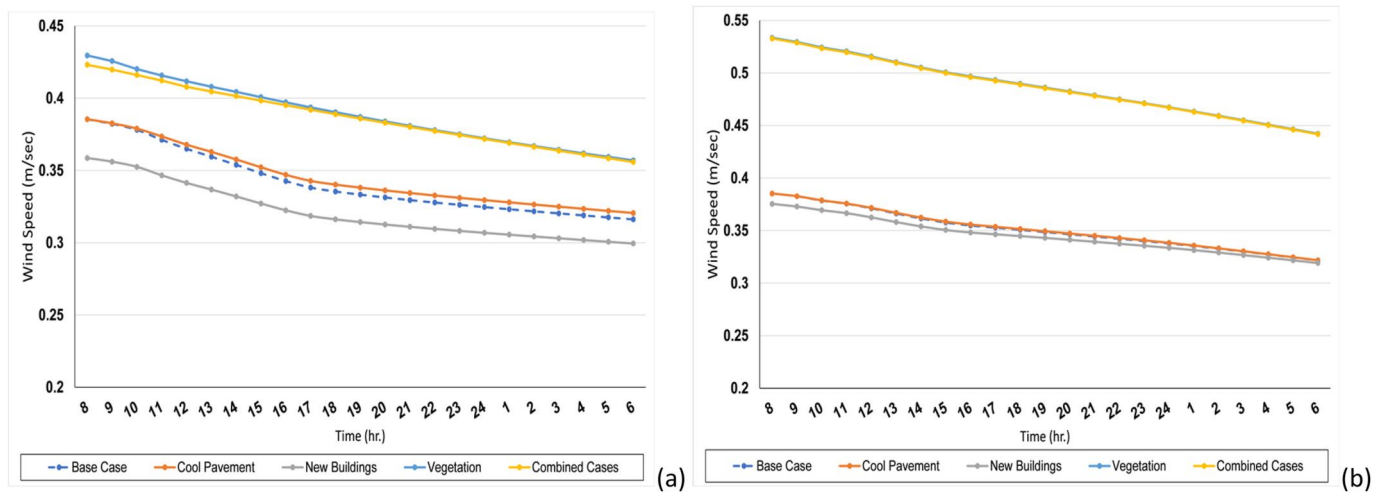


Fig. 8. Comparison of the average wind speed results between the base case scenario (BCS) and other mitigation scenarios during (a) 20th August and (b) January 20, 2018 for 24 h.

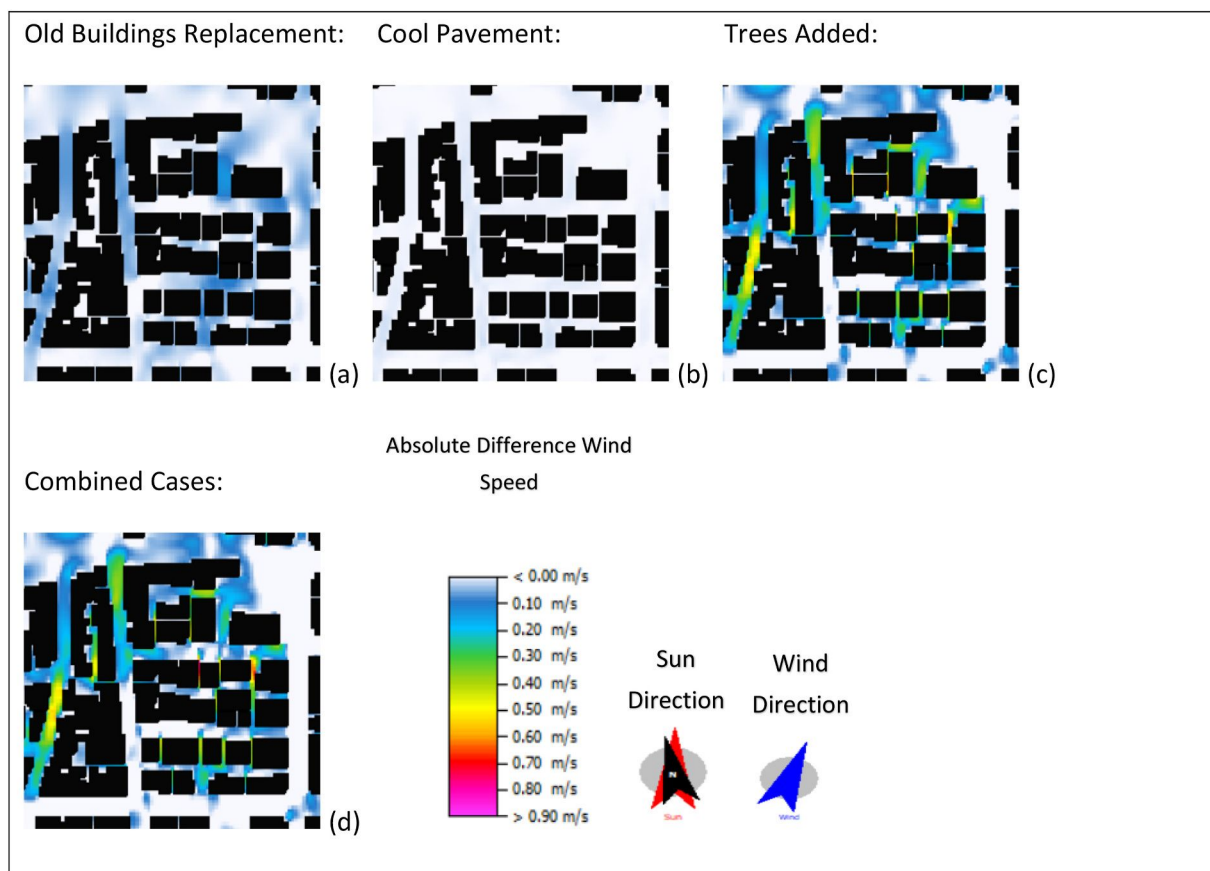


Fig. 9. Absolute difference in wind speed between various heat mitigation scenarios and the base case scenario at 12:00 at 1.4 m elevation from ground level.

5.5. Mean radiant temperature

Fig. 12 (a) displays the T_{mrt} simulation results. All the scenarios except the CPS have lower mean radiant temperature. The VS shows that the maximum difference reaches 10.5°C increase at noon, which is higher than the CCS due to the cool pavement.

Fig. 12(b) shows that the maximum difference between the scenarios is at noon while the difference is milder during the night. At noontime, the maximum difference is such that the BCS is 6.5°C higher than the

VS. There is a small difference between the CPS and the BCS, and no difference is reported when comparing the NBS to the BCS.

Fig. 13 shows the absolute difference in the T_{mrt} between the proposed scenarios and the BCS, at noon on the summer day. Given that the sun's location is due south, the NBS (Fig. 13 (a)) indicates that in particular places near the buildings, the difference is small compared to the BCS. The maximum difference of 20°C happens in the VS (Fig. 13 (c)), which is the highest of all the proposed cases, due to the shading provided by the trees. The CPS (Fig. 13 (b)) shows a 2°C increase in T_{mrt}

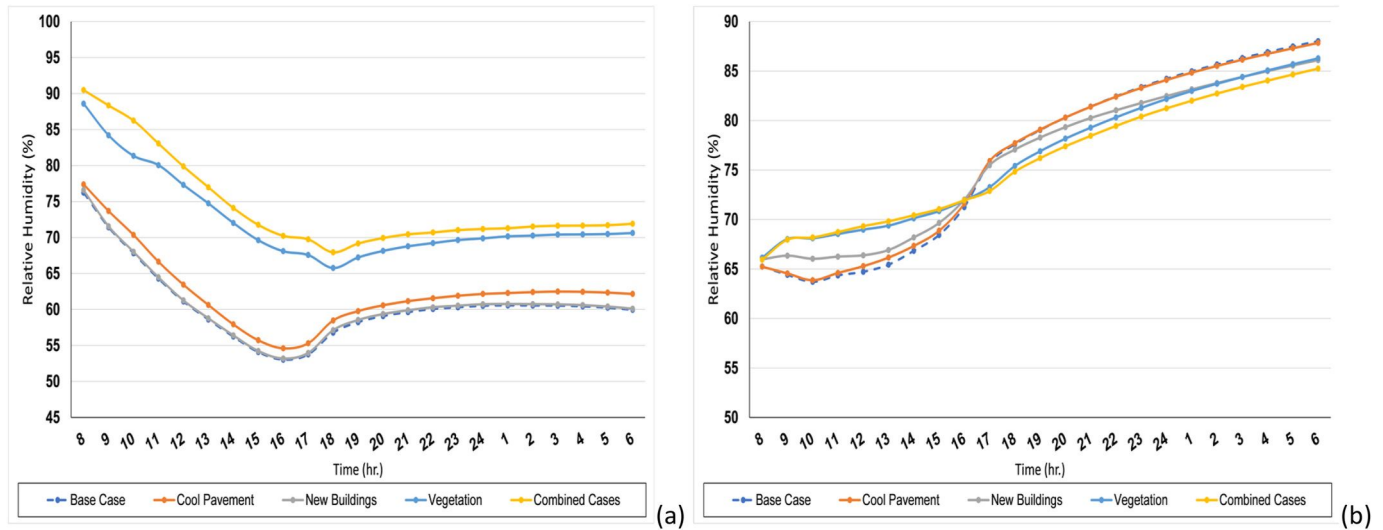


Fig. 10. Comparison of the average relative humidity results between the base case scenario and the other mitigation strategies during (a) 20th August and (b) January 20, 2018 for 24 h.

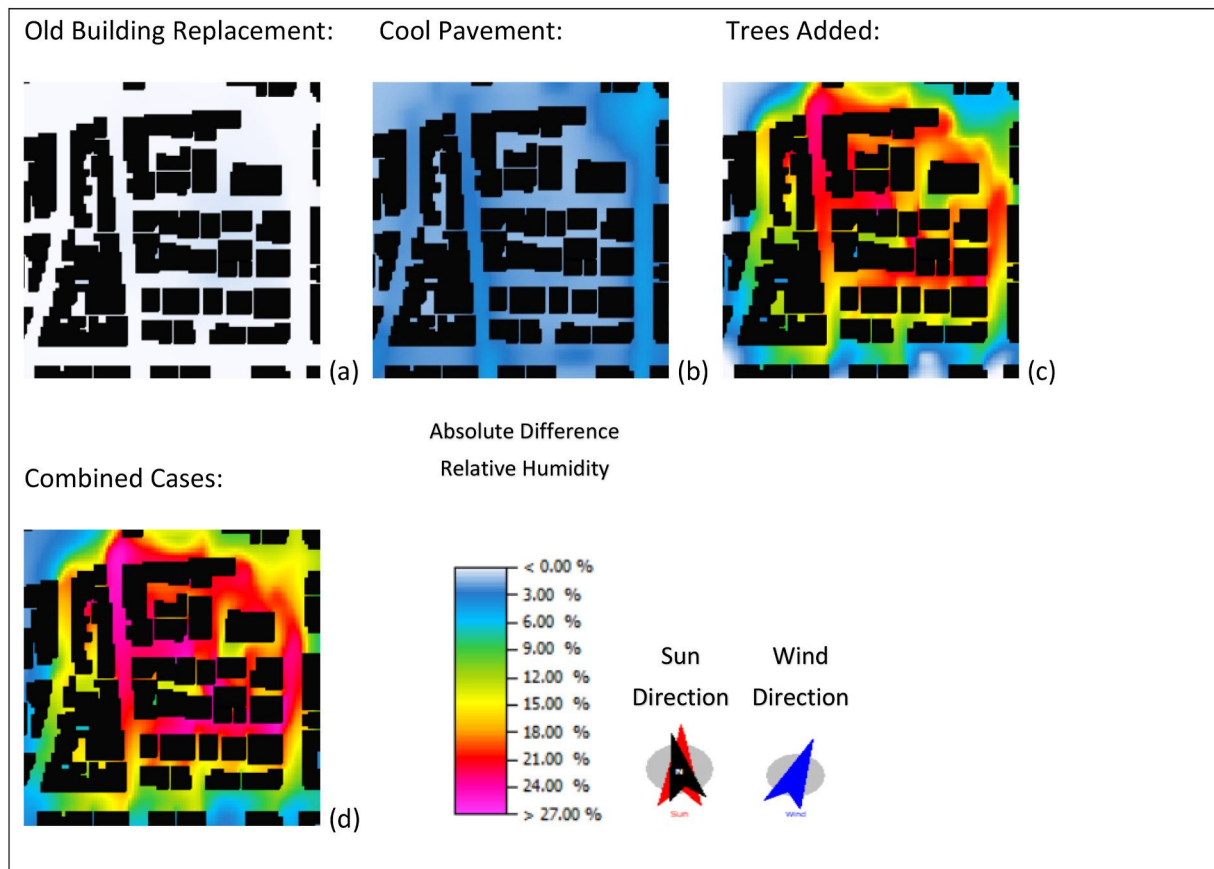


Fig. 11. Absolute difference in relative humidity between various heat mitigation scenarios and the base case scenario at 12:00 at 1.4 m elevation from ground level.

when compared to the BCS.

5.6. Ground surface temperature

Comparing the ground surface temperature to the base case model (Fig. 14 (a)) shows that the CPS, VS, and CCS cases have lower ground surface temperature during the day while the NBS displays no change. At noon, the maximum decrease of 6.5 °C in the ground surface

temperature happens in the CCS as compared to the BCS.

Fig. 14 (b) shows a small difference between the scenarios when compared to the BCS, with the maximum difference of 0.8 °C taking place in the CCS.

Fig. 15 presents the absolute difference in the ground surface temperature (T_g) between the proposed scenarios and the BCS at noon. The smallest effect among the proposed scenarios is the NBS (Fig. 15 (a)). Due to the solar altitude and azimuth, the maximum difference happens

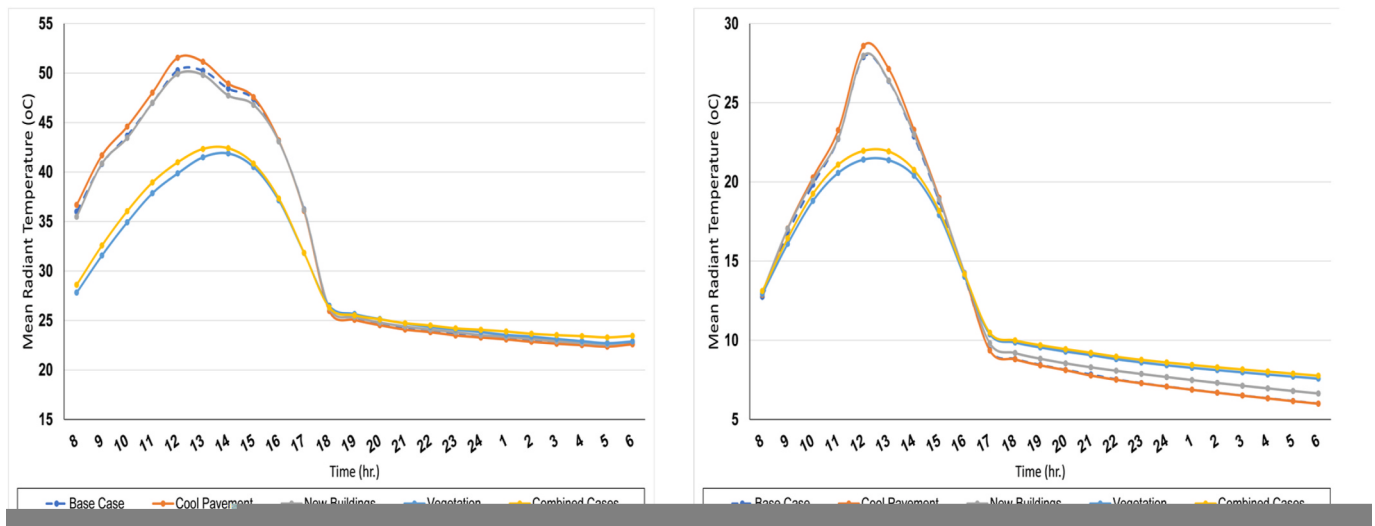


Fig. 12. Comparison of average mean radiant temperature simulation results between the base case model and other mitigation scenarios during (a) 20th August and (b) January 20, 2018 for 24 h.

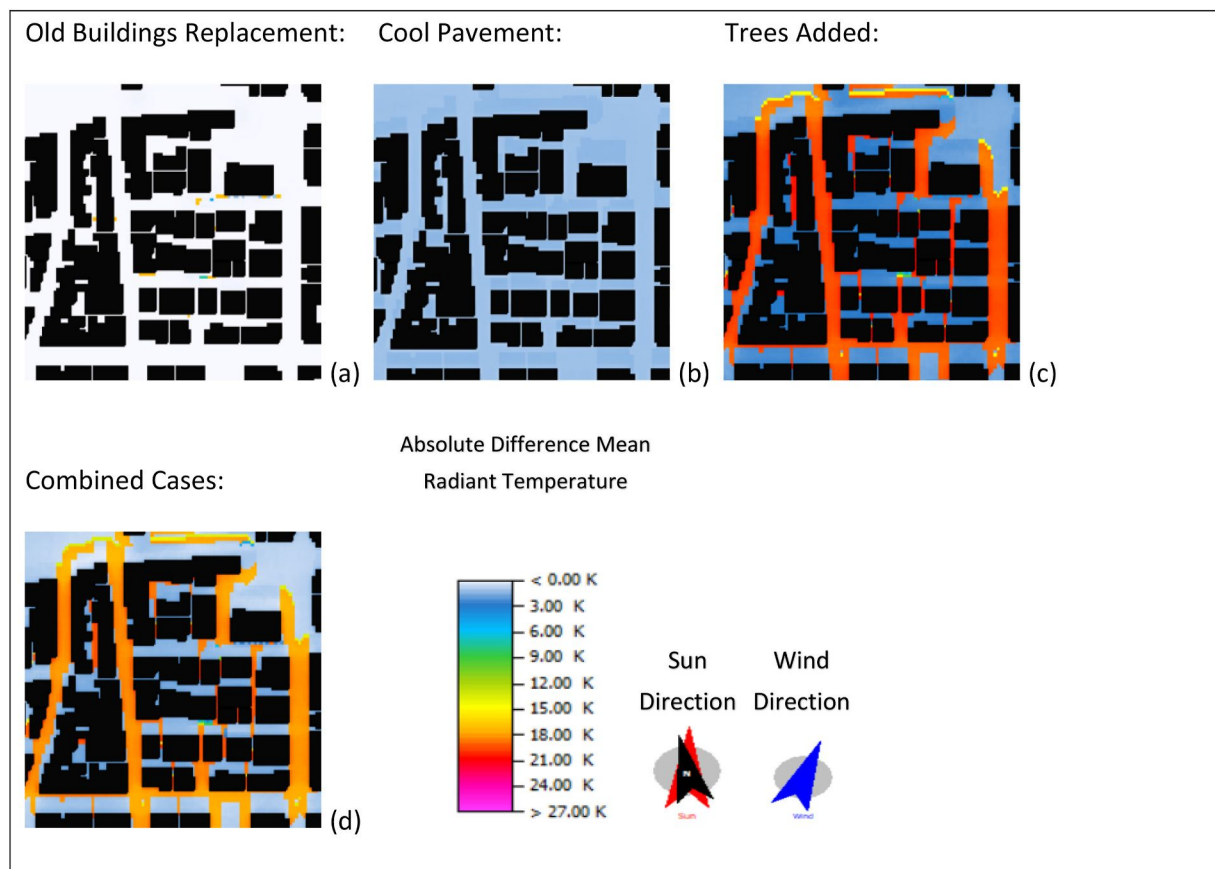


Fig. 13. Absolute difference in mean radiant temperature between various heat mitigation scenarios and the base case scenario at 12:00 at 1.4 m elevation from ground level.

on streets having a north-south orientation. When comparing the CCS to the BCS, the greatest difference (reduction in temperature) reaches 25 °C in some areas as shown in Fig. 15 (d). The decrease in the ground surface temperature is also in line with other studies whereby increasing the albedo by 0.25 leads to a decrease in the surface temperature by 10 °C [9].

6. Discussion

Given the climate of Beirut that is characterized by hot and humid summer months, a simulation for August 20 is conducted to discuss and highlight the most efficient mitigation strategies for the city. Introducing vegetation results in shading and evapotranspiration that contribute to decreasing the air temperature (2.6 °C) and providing a cooling effect.

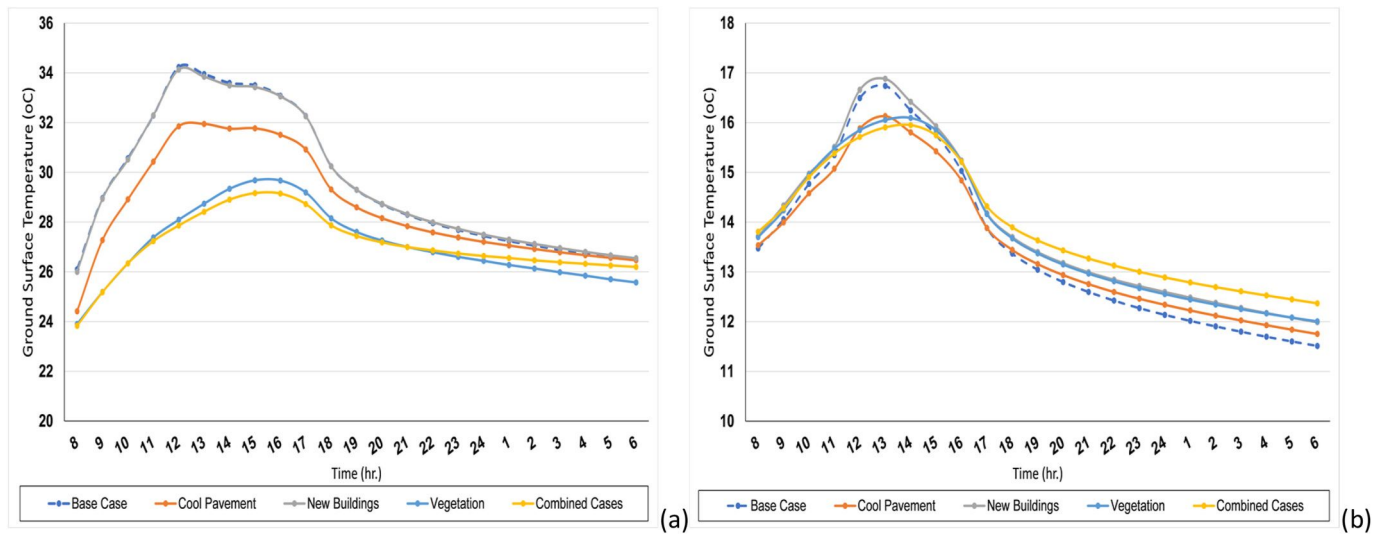


Fig. 14. Comparison of the average ground surface temperature results between the base case scenario and other mitigation scenarios during (a) 20th August and (b) January 20, 2018 for 24 h.



Fig. 15. Absolute difference in ground surface temperature between various heat mitigation scenarios and the base case scenario at 12:00 at the ground level.

This study reveals that after adding vegetation, the wind velocity has a very small increase of 0.07 m/s when compared to the BCS. This is shown as the average wind speed at the pedestrian level on a sidewalk. A closer look at the results shows that there is an inverse relationship between the relative humidity and the air temperature [51] whereby the VS has the highest relative humidity values with a difference of 15%. Adding vegetation will increase the shaded space, thus improving the thermal environment in an open outdoor space. Therefore, increasing

the urban vegetation in any area will protect the pedestrian from direct solar radiation (due to shading), resulting in reduced mean radiant temperature and air temperature, and thus enhancing the thermal comfort in the summer days [52].

Adding cool pavement shows a higher mean radiant temperature reaching 2 °C more than the BCS because of the increase in the albedo fraction of the ground cover (from 0.2 to 0.7); this is essentially due to the increase in the reflected radiation during the day. Even though, the

maximum decrease of 10.5 °C during the day in the mean radiant temperature is when comparing the BCS with the VS.

Adding new building material reveals that it does not have any effect on the ground surface temperature when compared to the BCS, while the cool pavement surface temperature is decreased by 4 °C because of the increase in the ground cover albedo, and a decrease by 6.5 °C when comparing the BCS with the VS due to the shading.

The simulation conducted for January 20, the coldest day of the year, reveals that the air temperature increases by an average of 0.7 °C throughout the whole simulation when comparing the BCS with the VS and the CCS, and the wind speed increases by 0.15 m/s. As for the relative humidity, the maximum difference is between the BCS and the VS. During the day and due to the evapotranspiration of the vegetation, the relative humidity increases by 5% and then decreases by 3% after sunset and during the night. The maximum mean radiant temperature difference of 6.5 °C is between the BCS and the VS, while, regarding the ground surface temperature, the maximum difference by 0.6 °C during the daytime is seen between the BC and the VS.

Therefore, the increase of vegetation in urban areas has a significant effect and is ultimately a better strategy for decreasing the air temperature and mean radiant temperature when compared to other mitigation strategies.

7. Conclusions

Urban areas with the appropriate thermal environment appeal to people. The climatic characteristics that affect the outdoor thermal environment are air temperature, wind speed, relative humidity, and mean radiant temperature. Identifying the most suitable definition of the LBC proved to be key to obtaining reliable results. Moreover, this research paper studied several mitigation strategies using the ENVI-met simulation software and showed the relative impact of each. This will help in enhancing the living conditions in urban areas. The results presented the importance of increasing the area of buildings on the periphery of the study area, which will contribute to numerical stability. This is particularly useful when there is a lack of measured data from weather stations located on the study site.

Based on the results, adding vegetation has a significantly positive impact on the thermal environment. As for the magnitude of the effect, vegetation is followed by the presence of cool pavement, and then by replacing old buildings with new construction. Vegetation protects pedestrians from incident solar radiation and reduces the air temperature due to transpiration. Increasing the vegetation cover would help protect pedestrians from direct solar radiation, thus reducing the mean radiant temperature and improving thermal comfort during summer. In addition, trees, by creating shade under the canopy result in the decrease of ground surface temperature.

Replacing old buildings with new ones and changing the materials of the constructions, shows a minor change in the mean radiant temperature because of the different densities of materials, their thermal conductivity, and specific heat capacity. Using reflective surfaces such as cool pavement with higher albedo absorbs less solar radiation and therefore has a lower surface temperature. Absorbing less solar radiation indicates that the cool pavement emits less longwave radiation, and the air temperature is reduced. Finally, the study revealed that combining the three proposed strategies has the most advantageous effect on improving the thermal comfort parameters in the summer.

In conclusion, this paper presents relevant guidelines that could support the design thinking process of architects, urban planners, and landscape designers. First, covering 26% of the pedestrian sidewalk with vegetation (canopy area) could significantly improve the thermal sensation. The second mitigation strategy consists of increasing the albedo of the ground cover and substituting the asphalt ground cover with cement concrete. Finally, the last mitigation strategy that showed the least impact on the assessment criteria was replacing the old buildings with new ones that include double wall and double glass windows with

different heights and materials.

The study identifies the considerable benefits that result from increasing vegetation in urban areas. This information can be considered and applied in other locations that have a climate similar to that of Beirut.

This work also provides a basis for further study and analysis regarding novel approaches in urban planning to mitigate the adverse effects of global warming in Mediterranean cities where adding vegetation should be considered as a practice that supports other mitigation strategies. Further investigation is required to integrate the behaviour of air pollution and its relationship to the mitigation strategies. The main limitation of this research is that the outputs and results rely on simulation software without any calibration because there were no stations in the studied area to compare with. Future research could also examine buildings' cooling energy load after adding vegetation as per the types of trees on the street level and/or on the rooftops. Furthermore, field measurements could be conducted to evaluate the simulated output data that would validate the strategies. Finally, anthropogenic heat release and car transportation heat fluxes could be accounted for in order to have a more comprehensive understanding of all the parameters that affect thermal comfort parameters in Beirut.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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